

SAI BALAJI R

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SUMMARY

M.Sc. Data Science student at VIT Chennai. I build AI software and backend systems — currently developing KnowledgeHUB-AI, a personal knowledge platform with semantic search, concept graphs, and RAG pipelines (milestone 8 of 12). Comfortable across the full stack: FastAPI, PostgreSQL, vector databases, Docker, and integrating LLMs into production software.

EXPERIENCE

Software Developer Intern · G2G Technologies · May 2024 – Jul 2024

- Worked on a live production School ERP (St. Ignatius Convent) — existing codebase, real users, real data.
- Owned the Attendance Management module end-to-end** — workflows, reports, and integration with existing ERP features across PHP, MySQL, and C#.

TECHNICAL SKILLS

Languages: Python, SQL, PHP, JavaScript, C#

Backend: FastAPI (working knowledge), Django, REST APIs, Node.js/Express

AI / ML: Scikit-learn, CNNs, YOLOv8, RAG, Vector Search, LLM Integration, Fuzzy Logic, Computer Vision

Databases: PostgreSQL, MySQL, Firebase, Qdrant, SQLite, Supabase

Tools: Git, Docker, Alembic, OpenCV, Streamlit, VS Code

PROJECTS

KnowledgeHUB-AI — AI Knowledge Platform · Python · FastAPI · Next.js · PostgreSQL · Qdrant · Docker · Alembic [\[Milestone 8/12\]](#) [↗ repo](#)

- Built from scratch — multi-user workspaces with JWT auth, 7+ format document ingestion (PDF, DOCX, PPTX, code, OCR, YouTube), and a concept graph linking every extracted idea back to its source chunk.
- Local-first RAG pipeline: Qdrant vector search + concept expansion + sufficiency scoring; every answer carries a provenance label. Full Docker stack, Alembic migrations, automated test coverage.

C-RAPS — Chennai Road Risk Analysis System · Python · YOLOv8 · Streamlit · Gemini API · scikit-fuzzy · OpenCV · GeoPandas [↗ repo](#)

- Real-time road safety platform — YOLOv8 vehicle + waterlogging detection feeds into a 6-input Mamdani fuzzy risk scorer; Gemini API generates citizen and official safety recommendations across 15 Chennai areas with live maps and weather integration.

Landroid — AI Land Intelligence Platform · FastAPI · React Native · TypeScript · Supabase/PostGIS · OpenCV · MapLibre [\[Hackathon\]](#) [↗ repo](#)

- Hackathon build (VIT Chennai × Birdscale 2026) — land health scoring and AI valuation using NDVI from Sentinel-2, soil pH from SoilGrids, and tree canopy detection from drone orthomosaics via OpenCV. FastAPI backend, Supabase+PostGIS, Firebase OTP auth.

E-Waste Safety — Hierarchical CNN + Fuzzy Risk Model · Python · CNN · Fuzzy Logic · Computer Vision [\[Conference Paper · ICGFG, VIT Chennai\]](#)

- Two-stage CNN for e-waste categorisation and lithium-ion battery detection to prevent pre-shredder fires; Mamdani fuzzy inference engine calibrated from thermal runaway physics with IEC 62061-compliant supervisory control. Co-authored with Niketha R & Dr. Balamurugan B J.

SJSSGJS School Fees ERP · Flutter · Django · PostgreSQL · REST APIs [↗ repo](#)

- Full-stack fee management system — Flutter mobile app consuming a Django REST backend, PostgreSQL handling fee structures, student records, and payment tracking.

K2 Essentials — Hostel Shop · JavaScript · Firebase Firestore [↗ repo](#)

- Browser-based ordering platform — student page + admin dashboard with real-time Firestore listeners; order status updates across both views instantly without page refresh.

Hadoop MapReduce vs Spark — Benchmark · PySpark · Hadoop · Python [↗ repo](#)

- Same VADER sentiment pipeline on 1.6M tweets. Spark: 11.2s vs MapReduce: 42s — 3.75× faster due to in-memory DAG execution vs HDFS disk writes between stages.

EDUCATION

M.Sc. Data Science (2025–2027, Pursuing) — VIT Chennai | CGPA: 8.56 / 10

B.Sc. Computer Science (2022–2025) — St. Xavier's College, Tirunelveli | CGPA: 8.08 / 10